

Harness metrics: graphify, the token ceilings, and the browser engines

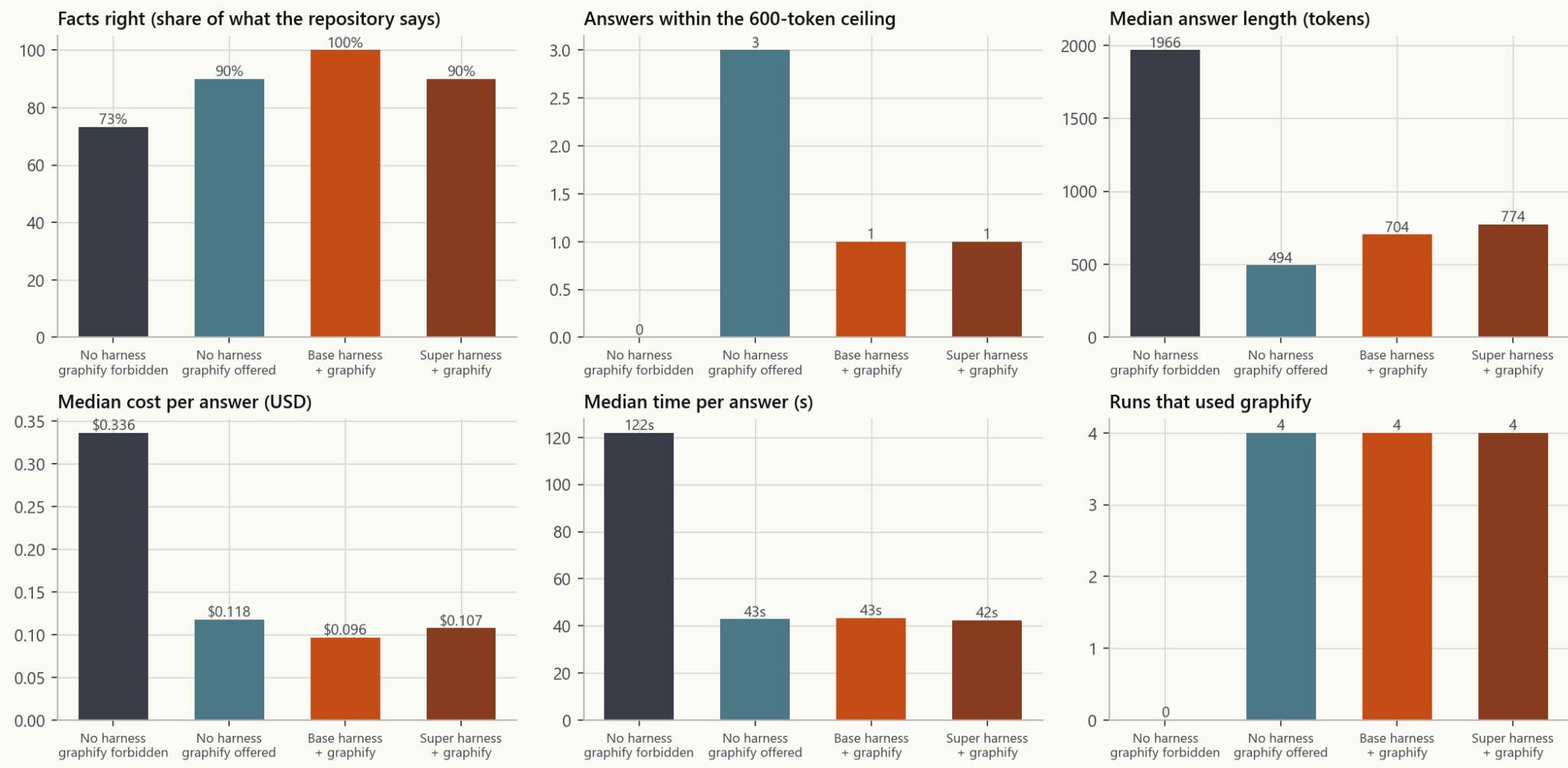
Every number here was measured on this machine and comes from a results file, not from an estimate. Token counts are the ones the API reported per response; cost is those counts at the published prices; time is first to last timestamp of the run. Highlighted rows are the harness arms.

1. The 600-token test, the one in benchmarks/run-it-yourself

Two questions, four arms, capped at 600 tokens. This is the test you can run yourself with the same prompts.

QUESTION	ARM	RUNS	FACTS RIGHT	WITHIN 600	MEDIAN ANSWER TOKENS	WORST ANSWER	MEAN COST	MEAN TIME	USED GRAPHIFY
complex	No harness, graphify forbidden	2	6/12	0/2	935	939	\$0.2812	82s	0/2
complex	No harness, graphify offered	2	12/12	1/2	762	966	\$0.1134	43s	2/2
complex	Base harness + graphify	2	12/12	1/2	589	658	\$0.1760	49s	2/2
complex	Super harness + graphify	2	12/12	1/2	1,140	1,746	\$0.1743	76s	2/2
super complex	No harness, graphify forbidden	2	16/18	0/2	1,294	1,803	\$0.3772	131s	0/2
super complex	No harness, graphify offered	2	18/18	1/2	556	612	\$0.2173	137s	2/2
super complex	Base harness + graphify	2	18/18	1/2	562	618	\$0.1521	53s	2/2
super complex	Super harness + graphify	2	18/18	0/2	740	812	\$0.2195	93s	2/2

The 600-token test: 16 runs, two questions, 11 answer(s) over the ceiling



The 600-token test: facts right, ceiling compliance, cost and time

1b. Does the ceiling hold? Rewriting the rule, measured

The first run showed answers going past a 600-token ceiling. Rule 15 was rewritten to state a word budget as well as a token count, and to ban the preamble and closing summary that usually blow a capped answer. The same sixteen runs were repeated. The rewrite made answers shorter and more accurate; it did not make the ceiling hold. Through the proxy the limit is a hard max_tokens and this question does not arise.

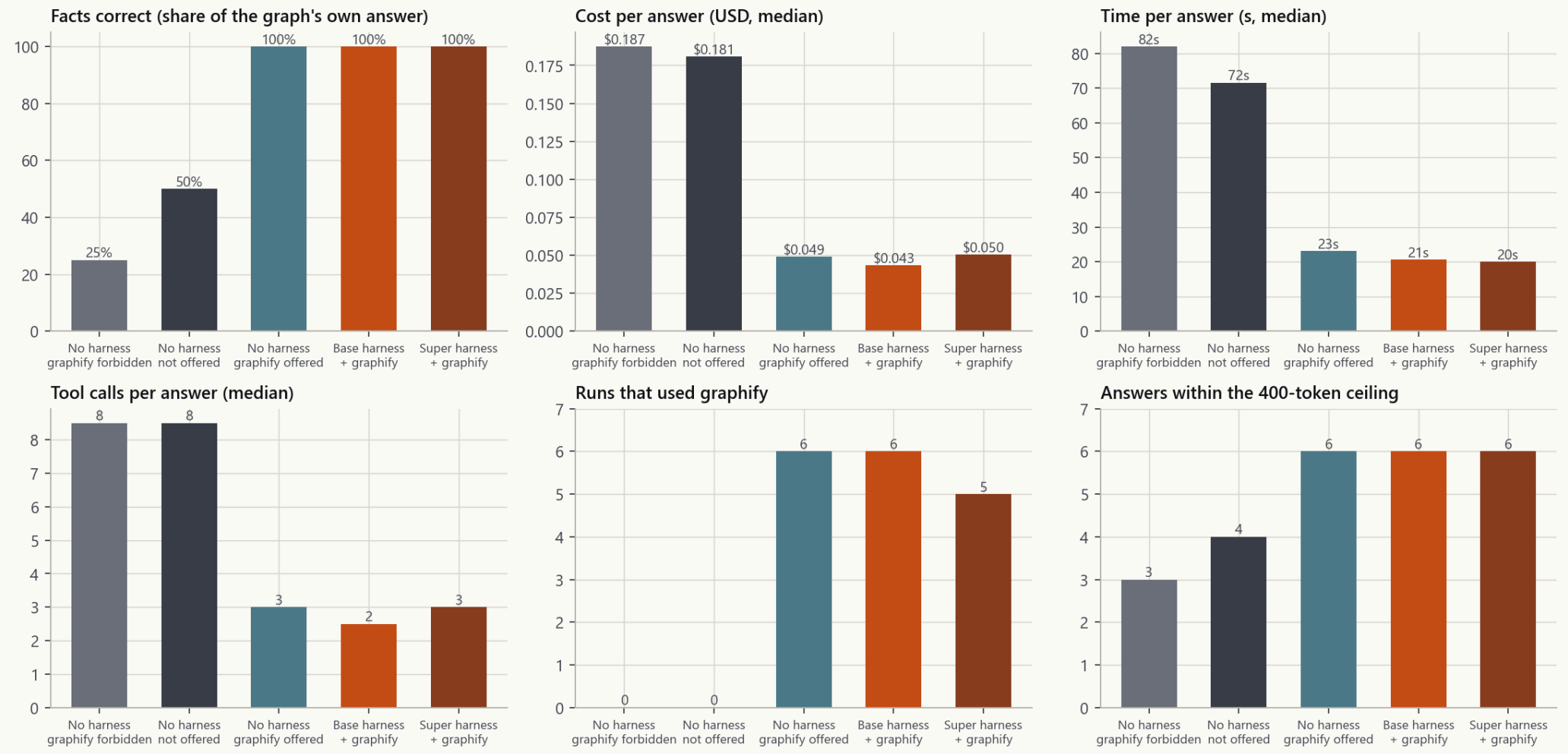
VERSION OF RULE 15	RUNS	WITHIN 600	MEDIAN ANSWER TOKENS	FACTS RIGHT	HARNESS ARMS WITHIN	HARNESS MEDIAN TOKENS
Rule stating the token ceiling only	16	5/16	720	106/120	2/8	704
Rule also stating a word budget, and banning preamble and summary	16	5/16	662	112/120	3/8	638

2. graphify across the harnesses, capped at 400 tokens

Five arms, two questions, three runs each. g1 is the question a graph answers; g2 is the control a grep answers.

QUESTION	ARM	RUNS	FACTS RIGHT	WITHIN 400	MEDIAN ANSWER TOKENS	WORST ANSWER	MEDIAN COST	MEDIAN TIME	USED GRAPHIFY
g1	No harness, graphify forbidden	3	0%	0/3	984	1,059	\$0.3918	158s	0/3
g1	No harness, not offered	3	33%	1/3	610	1,780	\$0.3311	156s	0/3
g1	No harness, graphify offered	3	100%	3/3	128	200	\$0.0535	21s	3/3
g1	Base harness + graphify	3	100%	3/3	268	270	\$0.0457	22s	3/3
g1	Super harness + graphify	3	100%	3/3	232	237	\$0.0572	25s	3/3
g2	No harness, graphify forbidden	3	100%	3/3	149	223	\$0.0473	13s	0/3
g2	No harness, not offered	3	100%	3/3	99	137	\$0.0435	12s	0/3
g2	No harness, graphify offered	3	100%	3/3	98	167	\$0.0346	25s	3/3
g2	Base harness + graphify	3	100%	3/3	162	269	\$0.0408	16s	3/3
g2	Super harness + graphify	3	100%	3/3	152	164	\$0.0272	13s	2/3

graphify across the harnesses: 30 runs, 2 codebase questions, 400-token ceiling (5 answer(s) over it)



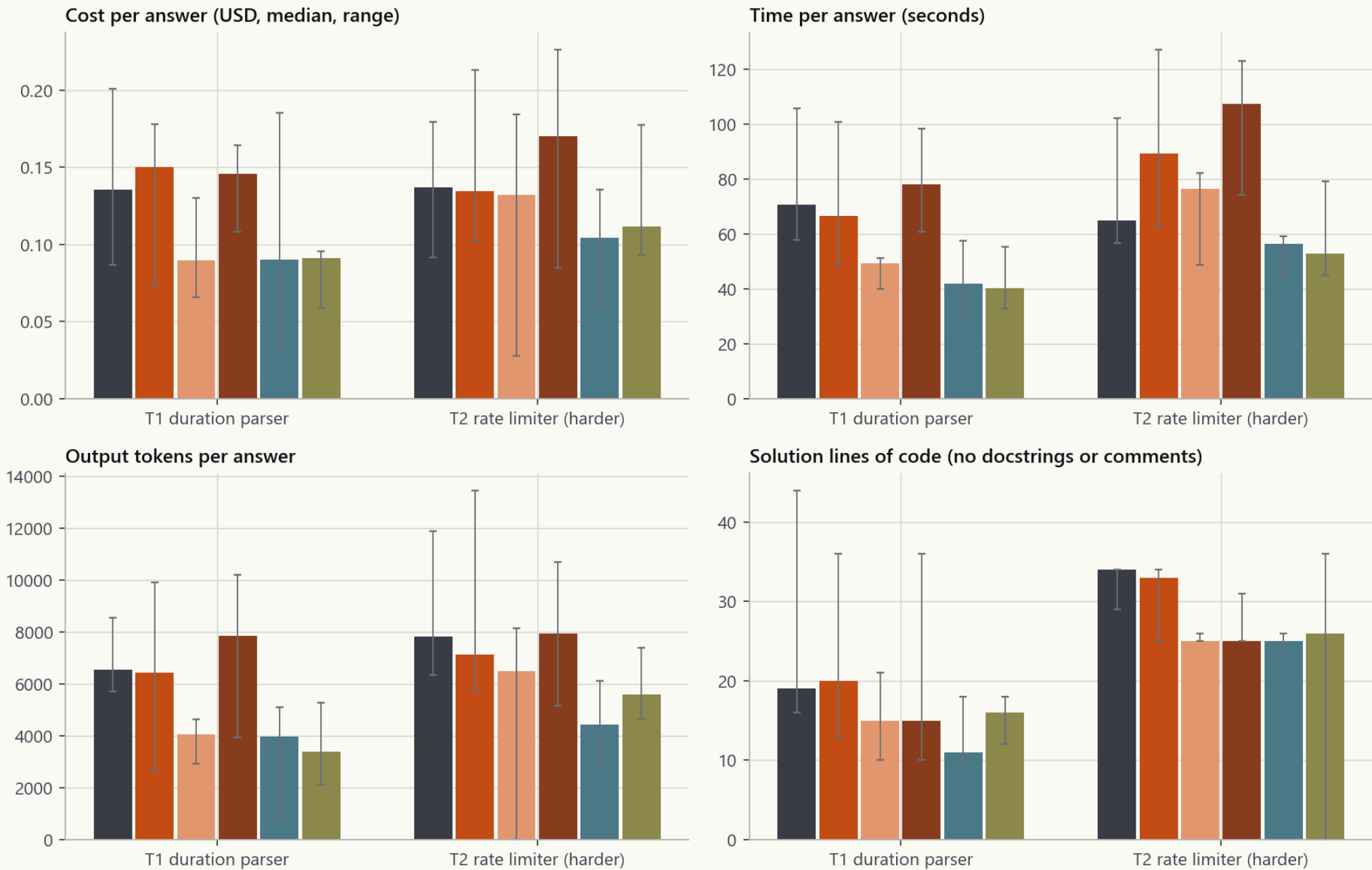
graphify across the harnesses, 30 runs at a 400-token ceiling

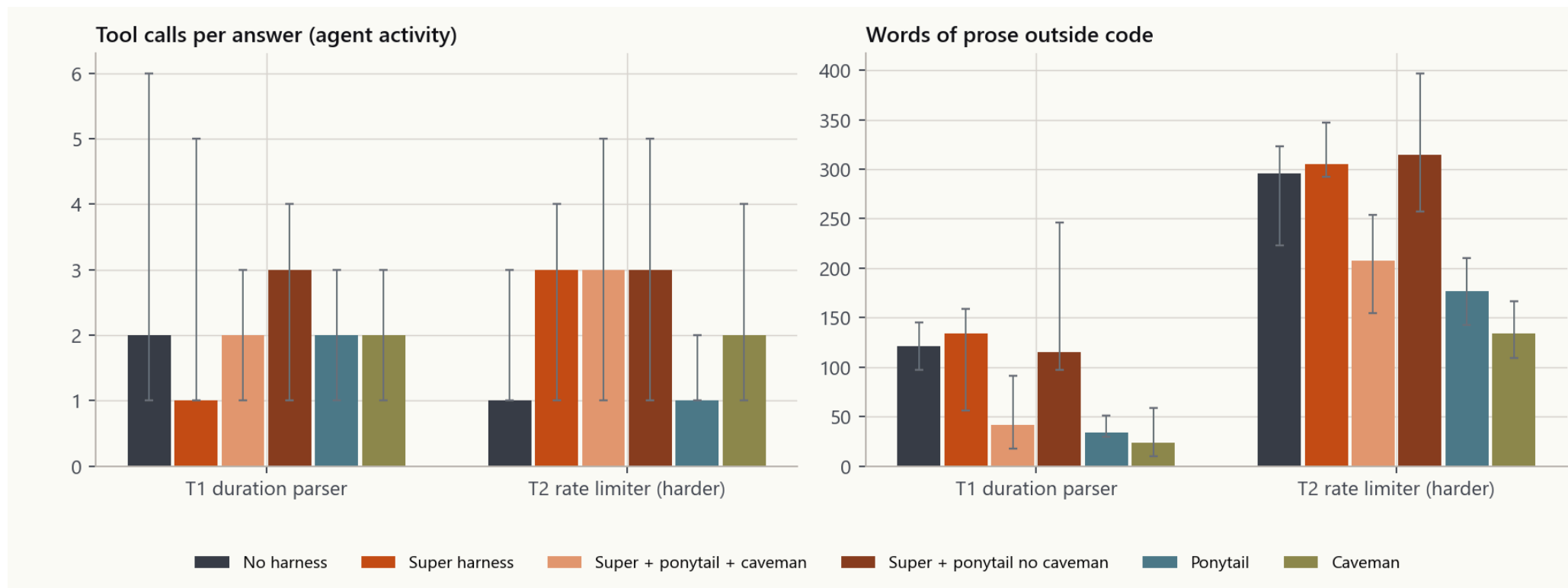
3. The harness and skill benchmark, 11 September

Six prompt arms, two coding tasks, five runs each, scored by hidden tests the model never saw and by two blind judges.

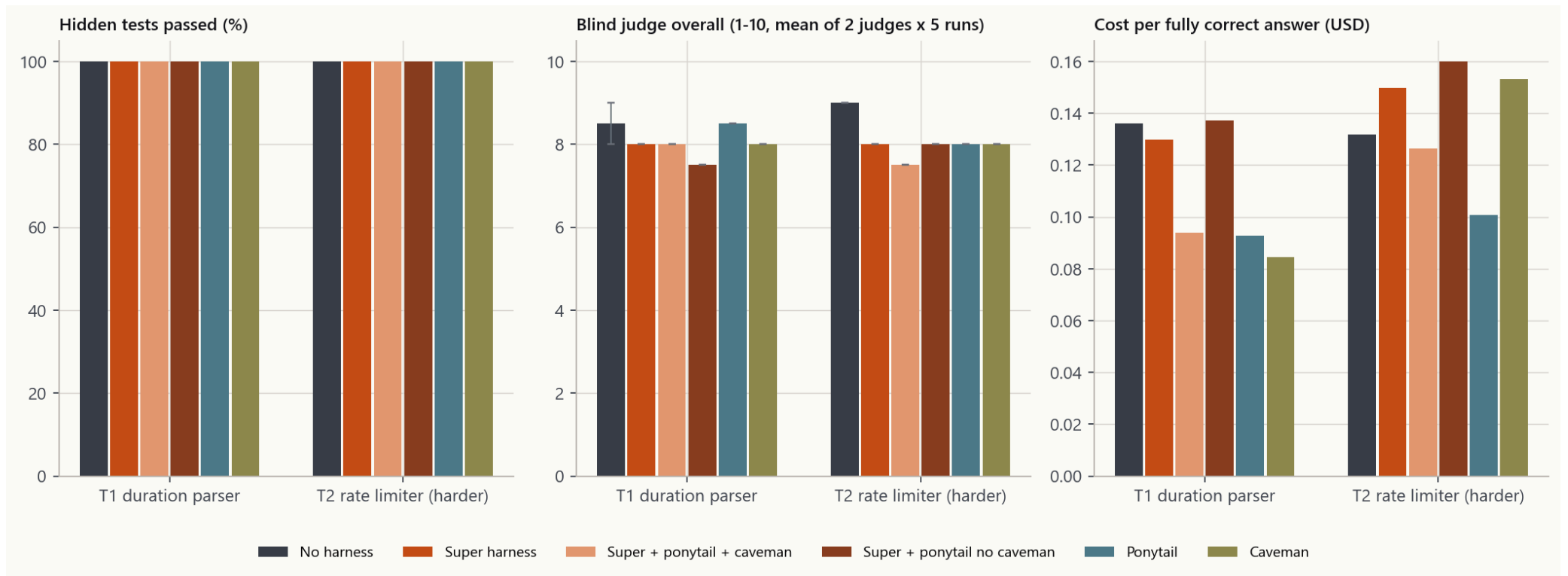
TASK	ARM	RUNS	COST	VS NO HARNESS	TIME	OUTPUT TOKENS	SOLUTION LOC	PROSE WORDS	HIDDEN TESTS	JUDGE /10
t1	No harness	5	\$0.1355	+0%	71s	6,534	19	121	100%	8.5
t1	Super harness	5	\$0.1500	+11%	67s	6,418	20	134	100%	8.0
t1	Super + ponytail + caveman	5	\$0.0895	-34%	49s	4,065	15	42	100%	8.0
t1	Super + ponytail, no caveman	5	\$0.1459	+8%	78s	7,840	15	115	100%	7.5
t1	Ponytail only	5	\$0.0900	-34%	42s	3,961	11	34	100%	8.5
t1	Caveman only	5	\$0.0913	-33%	40s	3,398	16	24	100%	8.0
t2	No harness	5	\$0.1372	+0%	65s	7,828	34	296	100%	9.0
t2	Super harness	5	\$0.1347	-2%	89s	7,115	33	305	100%	8.0
t2	Super + ponytail + caveman	5	\$0.1321	-4%	76s	6,477	25	208	100%	7.5
t2	Super + ponytail, no caveman	5	\$0.1704	+24%	108s	7,929	25	315	100%	8.0
t2	Ponytail only	5	\$0.1044	-24%	56s	4,415	25	177	100%	8.0
t2	Caveman only	5	\$0.1117	-19%	53s	5,575	26	134	100%	8.0

Coding benchmark: six prompt arms, two tasks, five runs each, Claude Sonnet 5





Six arms, two coding tasks, five runs each: cost, time, output, code size, tool calls and prose



Quality: hidden tests passed, blind judge score, and cost per fully correct answer

4. Browser engines: caveman-browse, Playwright, Rustwright

TASK	TOOL	RUNS	CORRECT	MEDIAN COST	MEDIAN TOKENS	MEDIAN TIME	CODE LINES	DECLINED BY SCAFFOLD
lookup	caveman-browse	3	3/3	\$0.1306	287,444	43s	5	0
lookup	playwright	3	3/3	\$0.0572	93,327	15s	13	0
lookup	rustwright	3	3/3	\$0.0596	94,410	22s	15	0
count	caveman-browse	3	3/3	\$0.1119	225,155	38s	3	0
count	playwright	3	3/3	\$0.0664	127,450	25s	34	0
count	rustwright	3	3/3	\$0.0780	128,697	59s	44	0
checkout	caveman-browse	2	2/2	\$0.3149	907,142	237s	16	0
checkout	playwright	2	2/2	\$0.1687	382,500	125s	31	0
checkout	rustwright	1	1/1	\$0.2387	617,578	100s	26	2

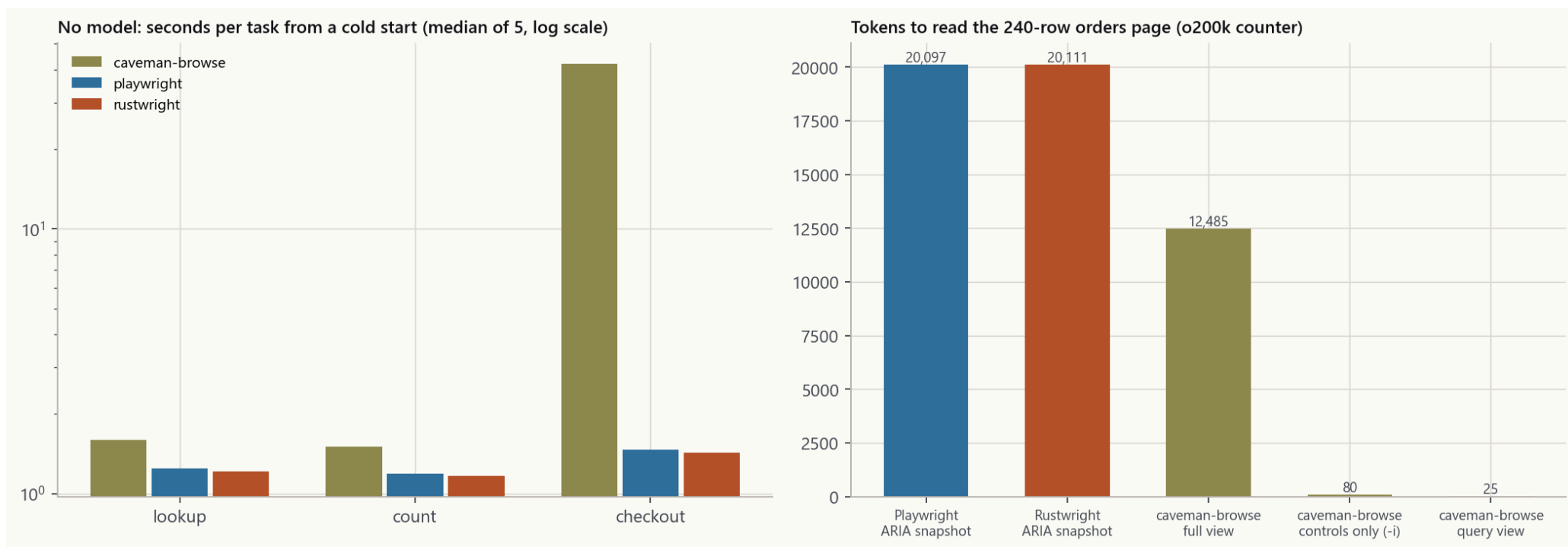
The same tasks with no model in the loop

TASK	TOOL	CORRECT	MEDIAN SECONDS	RANGE	CLIENT MIB
lookup	caveman-browse	5/5	1.59s	1.43 to 2.19	n/a
count	caveman-browse	5/5	1.50s	1.46 to 1.70	n/a
checkout	caveman-browse	5/5	42.13s	42.06 to 42.24	n/a
lookup	playwright	5/5	1.24s	1.15 to 3.39	174
count	playwright	5/5	1.19s	1.16 to 4.83	164
checkout	playwright	5/5	1.46s	1.44 to 1.63	197
lookup	rustwright	5/5	1.21s	1.16 to 1.40	52
count	rustwright	5/5	1.17s	1.09 to 1.29	51
checkout	rustwright	5/5	1.43s	1.34 to 1.49	52

What one page costs to read

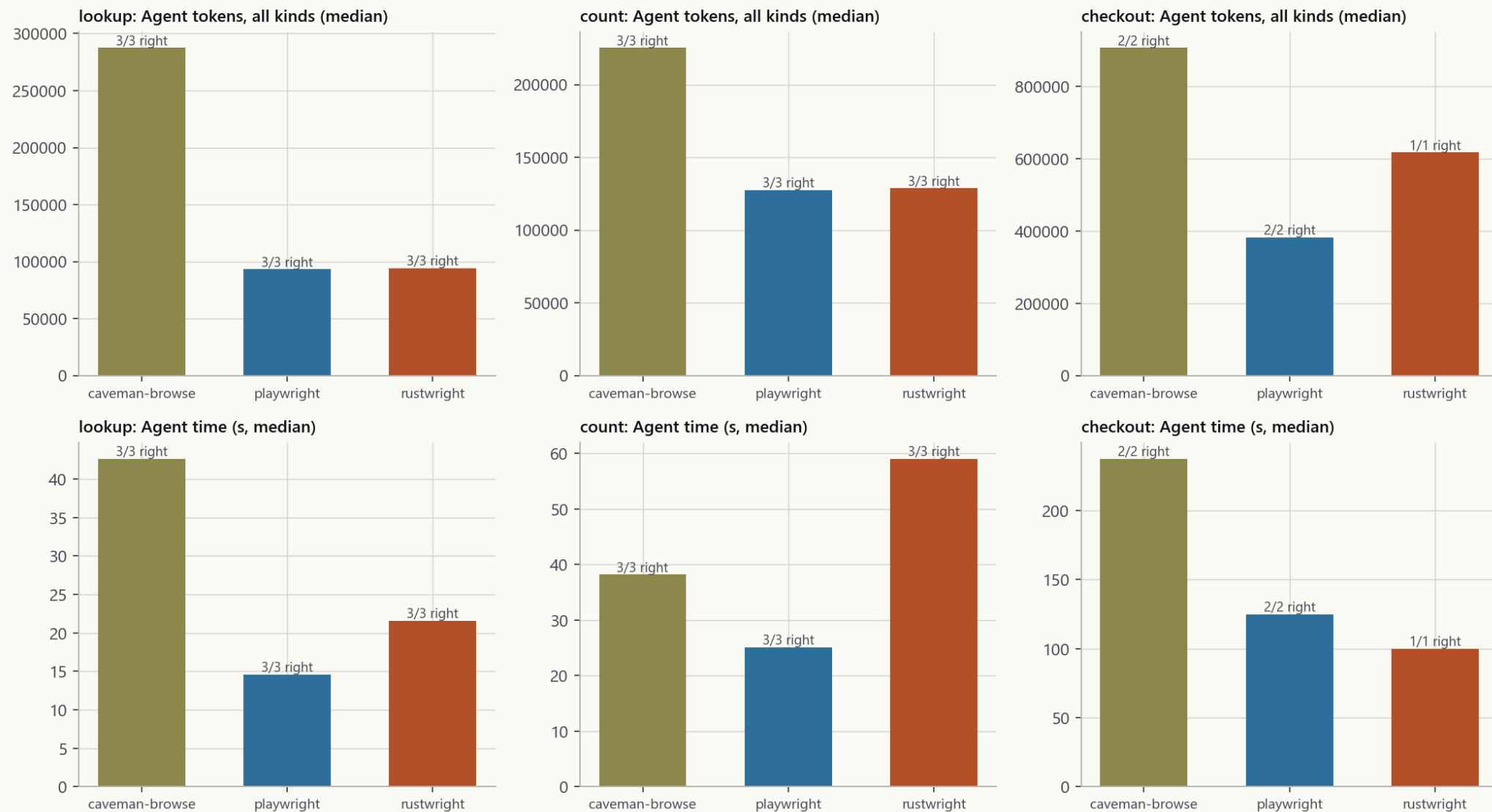
PAGE VIEW	BYTES	TOKENS (0200K)
caveman-browse:checkout:full	2,168	652
caveman-browse:checkout:interactive	296	80

PAGE VIEW	BYTES	TOKENS (0200K)
caveman-browse:orders:full	43,264	12,485
caveman-browse:orders:interactive	292	80
caveman-browse:orders:query	84	25
playwright:checkout:full	2,505	740
playwright:orders:full	63,820	20,097
rustwright:checkout:full	2,532	743
rustwright:orders:full	63,872	20,111



No model in the loop: seconds per task from a cold start, and tokens to read one page

Browser benchmark, agent in the loop (Claude Sonnet 5, three runs per tool per task)



An agent driving each tool: tokens and time, with how many runs got the right answer

Declined by scaffold counts runs that refused the task on read-only grounds without touching the tool. That is the agent profile used for isolation, not a property of the browser tool, and those runs are excluded from the medians rather than counted as failures.